Natural Language Inference: Is type theory relevant?

Stergios Chatzikyriakidis (based on joint work with Zhaohui Luo)

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Natural Language Inference

- Roughly: the task of determining whether a NL hypothesis H follows from an NL premise P
  - 'inferential ability is not only a central manifestation of semantic competence but is in fact centrally constitutive of it' [Cooper et al. 1996]
  - Inferential ability as a means to test the semantic adequacy of NLP systems

Natural Language Inference

- But NLI has been proven to be a notorious difficult problem
  - Shallow and Deep approaches
  - Translation into an intermediate logical language or not
    - ★ Mostly expressively weak logical systems, i.e. first order logic in deep approaches [Bos and Markert(2005), Pulman(2013), Chatzikyriakidis and Luo(2014)]
    - ★ Various shallow approaches, e.g. bag of words, machine learning methods [Romano et al.(2006)Romano, Kuylekov, Szpektor, Dagan, and Lavelli, Glickmann et al.(2005)Glickmann, Dagan, and Koppel, Hickl et al.(2005)Hickl, Williams, Bensley, Roberts, Rink, and Shi, MacCartney et al.(2008)MacCartney, Galley, and C.D. Manning]
    - ★ Recently, approaches using distributional vector spaces, deep learning methods (see for example [Bowman et al.(2015)Bowman, Angeli, Potts, and Manning, Wang and Jiang(2015)])
    - ★ Hybrid approaches (MacCartney 2008)

Natural Language Inference

- What counts as an NLI?
  - Logical Entailment
    - The FraCaS test [Cooper and Ginzburg(1996)] suite provides a collection of mostly logical entailments. Categorization is done according to semantic category
    - Three way classification of 346 inference problems: YES (the conclusion follows), NO (the negation of the conclusion follows) and UNK (none of the two follow)

(1) A Swede won the Nobel Prize.
Every Swede is Scandinavian.
Did a Scandinavian win the Nobel prize? [Yes, FraCas 049]

(2) No delegate finished the report on time..
Did any Scandinavian delegate finish the report on time? [No, FraCaS 070]

Natural Language Inference

(3) A Scandinavian won the Nobel Prize.
Every Swede is Scandinavian.
Did a Swede win the Nobel prize? [UNK, FraCaS 065]

- Other typical examples

(4) Either Smith, Jones or Anderson signed the contract. Did John sign the contract? [UNK] (plurals, FraCaS 083)

(5) Dumbo is a large animal. Is Dumbo a small animal? [NO] (adjectives, FraCaS 205)

(6) Smith believed that ITEL had won the contract in 1992. Did ITEL win the contract in 1992? [UNK] (Attitudes, FraCaS 334)

The FraCaS: Brief evaluation

- Unnatural data
  - Highly skilled formal semanticists provided types of examples that, according to them, an NLI system should be able to capture
  - Small dataset for non-logical approaches. For modern day systems like deep learning systems, such an amount of data is out of the question
- However the FraCaS is a useful resource since it contains targeted examples according to linguistic phenomenon
  - Involves cases of very fine-grained inference that the newer platforms do not have (e.g. reasoning with elliptical fragments, aspectual inference etc.)

The RTE platforms

- Platform appeared in 2005
  [Dagan et al.(2006)Dagan, Glickman, and Magnini]
  - New version every year until 2011 (RTE7)
  - RTE concentrates on real rather than constructed data
  - A natural piece of text is taken as a premise and then a hypothesis is constructed out of it
    - ★ The first RTEs had two way entailment (entailment, non-entailment), three way classification was added in the later suites

Some RTE examples

P Budapest again became the focus of national political drama in the late 1980s, when Hungary led the reform movement in eastern Europe that broke the communist monopoly on political power and ushered in the possibility of multiparty politics.

H In the late 1980s Budapest became the center of the reform movement. [follows, RTE702]

Some RTE examples

P Budapest again became the focus of national political drama in the late 1980s, when Hungary led the reform movement in eastern Europe that broke the communist monopoly on political power and ushered in the possibility of multiparty politics.

H In the late 1980s Budapest became the center of the reform movement. [follows, RTE702]

P Like the United States, U.N. officials are also dismayed that Aristide killed a conference called by Prime Minister Robert Malval in Port-au-Prince in hopes of bringing all the feuding parties together.

H U.N. officials take part in a conference called by Prime Minister Robert Malval. (does not follow, RTE1933)

The RTE platform: Brief evaluation

- Natural examples
- Inclusion of non-logical inference cases'
  - Cases where vast amounts of world knowledge has to be taken into consideration
- Complicated in terms of their syntax and semantics given the open text nature but:
  - Simple at the same time, given that in most of the cases no deep reasoning is involved in the examples (at least the kind discussed by formal semanticists)
- This surface complexity is evident in the poor performance of all systems evaluated on the RTE tasks
- No distinction between various types of inference (logical, presuppositional, enthymematic etc. (this is to a extent a drawback of the FraCaS))
- For models that rely on big training sets, RTE is extremely small a dataset (less than 1000 cases for each suite)

The SNLI platform

- Developed at Stanford
  [Bowman et al.(2015)Bowman, Angeli, Potts, and Manning]
  - Created using crowdsourcing (mechanical turk)
  - The subjects are given the caption of a picture and are asked to provide:
    - ★ An alternate true caption
    - ★ An alternate possibly true caption
    - ★ An alternate false caption

The SNLI platform

- Instructions used on Mechanical Turk

We will show you the caption for a photo. We will not show you the photo. Using only the caption and what you know about the world:

- Write one alternate caption that is **definitely** a **true** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "There are animals outdoors."*
- Write one alternate caption that **might be** a **true** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "Some puppies are running to catch a stick."*
- Write one alternate caption that is **definitely** a **false** description of the photo. *Example: For the caption "Two dogs are running through a field." you could write "The pets are sitting on a couch." This is different from the* maybe correct *category because it's impossible for the dogs to be both running and sitting.*

The SNLI platform

- Huge dataset
  - Good for training NN models
- Inference examples are grounded in situations
- Validation: subset of the sentences were sent to a validation task
- Misses a number of more generic inferences
  - Everything is tied to situations
  - For example: what would be the caption of *all men are human*?
- Again, it seems that this platform also misses parts of the whole range of phenomena associated with NLI
- On a theoretical basis, a platform that captures the complexity and coverage of NLI is not yet to be found
  - We are currently thinking about this along with Robin Cooper and Simon Dobnik but I will say no more here

A brief intro to Modern Type Theories (MTTs)

- Type Theories within the tradition of Martin Löf
  - In linguistics, this work has been initiated by pioneering work of Ranta (1994)
- Here, we use one such MTT, UTT, first applied by Luo (2010) to the study of linguistic semantics
  - Two characteristics that are promising in using MTTs as an alternative formal semantics language:
    - ★ Consistent internal logic according to the propositions-as-types principle
    - ★ Rich type structures

Intro to MTTs-Type Many Sortedness and Rich Typing

- Many-sortedness of types
  - Use of many types to interpret CNs, *man* and *table*
  - CNs are interpreted as Types rather than as predicates ($e \rightarrow t$)
- Use of Dependent Types Π and Σ
  - When A is a type and P is a predicate over A, $\Pi x{:}A.P(x)$ is the dependent function type that stands for the universally quantified proposition $\forall x{:}A.P(x)$
  - Π for polymorphic typing: $\Pi A{:}CN.(A \rightarrow Prop) \rightarrow (A \rightarrow Prop)$
  - A is a type and B is an A-indexed family of types, then $\Sigma x{:}A.B(x)$, is a type, consisting of pairs (a, b) such that a is of type A and b is of type $B(a)$.
  - Adjectival modification as involving Σ types (Ranta, 1994; Luo, 2010): $[\![heavy_book]\!] = \Sigma x : book.heavy(x)$

Intro to MTTs-Subtyping

- Coercive subtyping
  - Can be seen as an abbreviation mechanism
    - ★ A is a (proper) subtype of B ($A < B$) if there is a unique implicit coercion c from type A to type B
    - ★ An object a of type A can be used in any context $\mathfrak{C}_B[_]$ that expects an object of type B: $\mathfrak{C}_B[a]$ is legal (well-typed) and equal to $\mathfrak{C}_B[c(a)]$.
    - ★ For example assuming $man < human$, $John : man$ and $shout : human \to Prop$, then $shout(John)$ is well-typed.

Intro to MTTs-Universes

- Universes
  - A universe is a collection of (the names of) types into a type (Martin Löf, 1984).
  - Universes can help semantic representations. For example, one may use the universe $\mathrm{CN} : \mathit{Type}$ of all common noun interpretations and, for each type A that interprets a common noun, there is a name $\overline{A}$ in CN. For example,

    $$\overline{\mathit{man}} : \mathrm{CN} \quad \text{and} \quad T_{\mathrm{CN}}(\overline{\mathit{man}}) = \mathit{man}.$$

    In practice, we do not distinguish a type in CN and its name by omitting the overlines and the operator T_{CN} by simply writing, for instance, $\mathit{man} : \mathit{CN}$.

Modification

- Adjectival modification as involving Σ types, in line with Ranta (1994)
  - Intersective adjectives as simple predicate types and subsective as polymorphic types over the CN universe:
    - ⋆ $[\![black]\!] : Object \rightarrow Prop$
    - ⋆ $[\![small]\!] : \Pi A{:}CN.A \rightarrow Prop$ (the A argument is implicit)
    - ⋆ For *black man*, we have: $\Sigma m{:}\,[\![man]\!]\,.\,[\![black]\!](m) < [\![man]\!]$ (via π_1)
    - ⋆ $< \Sigma m{:}\,[\![human]\!]\,.\,[\![black]\!](m)$ (via subtyping propagation)
    - ⋆ $< [\![human]\!]$ (via π_1)
  - For *small man*:
    - ⋆ $\Sigma m{:}\,[\![man]\!]\,.\,[\![small]\!]\,[\![man]\!](m) < [\![man]\!]$ (via π_1)
    - ⋆ BUT NOT:
      $\Sigma m{:}\,[\![man]\!]\,.\,[\![small]\!]\,[\![man]\!](m) < \Sigma m{:}\,[\![animal]\!]\,.\,[\![small]\!]\,[\![man]\!](m)$
    - ⋆ Many instances of small: $small([\![man]\!])$ is of type $[\![man]\!] \rightarrow Prop$, $small([\![animal]\!])$ is of type $[\![animal]\!] \rightarrow Prop$

Adjectival Modification/More Advanced Issues

- Privative adjectives like *fake*
  - We follow Partee (2007) and argue that privative adjectives are actually subsective adjectives which operate on CNs with extended denotations
    - ★ For exaple, the denotation of *fur* is expanded to include both *real* and *fake* furs:

      (7) I don't care whether that fur is fake fur or real fur.

      (8) I don't care whether that fur is fake or real.

    - ★ $G = G_R + G_F$ with $inl(r){:}G_R$ and $inl(f){:}G_F$
    - ★ Injections as coercions: $G_R <_{inl} G \texttt{and} G_F <_{inr} G$ and we define: $real_gun(inl(r)) = True \quad and \quad real_gun(inr(f)) = False$; $fake_gun(inl(r)) = False \quad and \quad fake_gun(inr(f)) = True$.
- Non-committal adjectives like *alleged*
  - Use of TT contexts representing beliefs (Ranta 1994) $[\![alleged\ N]\!] = \Sigma p{:}Human.\ B(p, A_N)$

Adjectival Modification/More Advanced Issues

- Dealing with additional parameters: grades, temporal arguments
  - Use of indexed types
    - ★ Basically, CNs with indexed arguments
    - ★ For example, in order to reason about *height* in *George is 1.60 tall*, one needs to be able to refer to a *height* parameter
    - ★ We define type $[\![Human]\!] : Height \rightarrow CN$
    - ★ $Human_i\ (i{:}Height)$ stands for humans indexed by i.
    - ★ Gradable adjectives are defined as taking indexed CN arguments (e.g. $[\![short]\!] : Human_i \rightarrow Prop$)
  - Different degree parameters are needed (e.g. *height,size,width* or even abstract ones like *idiocy* (for example in *he is a huge idiot*))
    - ★ Introduce a universe of degrees (D) that will contain all degree types
    - ★ All types in the universe are totally ordered, anti-symmetric, reflexive and dense

Adjectival Modification/An example: *tall*

- We first use the auxiliary object *TALL* and then define *tall* to be its first projection, π_1
  - $SHORT : \Pi i : Height.\Sigma p : Human(i) \rightarrow Prop.\forall h_1{:}Human(i).$ $p(h_1) \leftrightarrow i < n$
  - $[\![short]\!](i) = \pi_1(SHORT(i)) : Human(i) \rightarrow Prop$
- n is a contextual parameter, the standard value provided by the context
- $[\![STND]\!] = \lambda A{:}\mathrm{CN}.\lambda i{:}D.\lambda P{:}A_i \rightarrow Prop.\exists n_1{:}Nat.n_1 = n \wedge i <> n$
  - *short* basically returns the first component of the pair $SHORT(i)$ of type $Human_i \rightarrow Prop$
  - The inference *John is tall* $\Rightarrow$ *John is taller than the standard value* follows from the second component of $SHORT(i)$
    - Assuming that $tall(John_i)$ is $p(h_1)$, $i < n$ follows
    - Similar account for comparatives: Instead of a relation between an i and the standard, we have a relation between i and j provided by two arguments $Human_i$ and $Human_j$

Adjectival Modification/Multidimensional Adjectives

- Quantification across different dimensions
  - E.g. to be considered *healthy* one has to be healthy w.r.t a number of dimensions (blood pressure, cholesterol etc.)
    - ★ Involves universal quantification over dimensions
  - The antonyms of these type of multidimensional adjectives existentially quantify over dimensions
    - ★ For one to be sick, only one dimension is needed
- We formulate this idea by Sassoon (2008) as follows:
  - We define an inductive type *health*
    - ★ $\mathit{Inductive}\ [\![\mathit{Health}]\!] : D := \mathit{Heart} \text{ — } \mathit{Blood_pressure} \text{ — } \mathit{Cholesterol}$
  - Then we define:
    - ★ $[\![\mathit{healthy}]\!] = \lambda x{:}\mathit{Human}.\forall h{:}\mathit{Health}.\mathit{Healthy}(h)(x)$
    - ★ $[\![\mathit{sick}]\!] = \lambda x{:}\mathit{Human}.\neg(\forall h{:}\mathit{Health}.\mathit{Healthy}(h)(x))$

Adverbial Modification

- Typing issues: How are we going to type adverbs in a many sorted TT?
  - Two basic types
    - ★ Sentence adverbs: $Prop \rightarrow Prop$
    - ★ VP-adverbs: $\Pi A{:}CN.(A \rightarrow Prop) \rightarrow (A \rightarrow Prop)$
    - ★ Polymorphic type: Depends on the choice of A
    - ★ Given that we are talking about predicates, depends on the choice of the argument
    - ★ $[\![walk]\!] : Animal \rightarrow Prop \Rightarrow [\![ADV]\!]\,[\![walk]\!] : (Animal \rightarrow Prop)$
    - ★ $[\![drive]\!] : Human \rightarrow Prop \Rightarrow [\![ADV]\!]\,[\![drive]\!] : (Human \rightarrow Prop)$

Adverbial Modification: Veridicality

- Veridical Adverbials when applied to their argument, imply their argument
  - John opened the door quickly $\Rightarrow$ John opened the door
  - Fortunately, John is an idiot $\Rightarrow$ John is an idiot
- Non-veridical adverbs do not have this property
  - John allegedly opened the door $\not\Rightarrow$ John opened the door

Adverbial Modification: Veridicality

- We can use a similar organizational strategy as in the case with adjectives
  - Define an auxiliary object first, define the adverb as its first projection
    - ★ $[\![VER_{Prop}]\!] : \Pi v : Prop.\ \Sigma p : Prop.p \supset v$
    - ★ $[\![ADV_{ver-Prop}]\!] = \lambda v : Prop.\ \pi_1(VER_{Prop}(v))$
  - An adverb like *fortunately* will be defined:
  - $[\![fortunately]\!] = \lambda v : Prop.\ \pi_1(VER_{Prop}(v))$
- Consider the following: Fortunately, John went $\Longrightarrow$ John went
  - The second component of $(VER_{Prop}(v))$ is a proof of $[\![fortunately]\!](v) \Rightarrow v$
  - Taking v to be $[\![John\ went]\!]$, the inference follows

Adverbial Modification: Intensional/domain adverbials

- Use of TT contexts in this case as well
  - $[\![allegedly]\!] = \lambda P : Prop.\ \exists p{:}\ [\![human]\!], B_p(P)$
  - Someone has alleged that P (P is an agent's belief context (Chatzikyriakidis 2014; Chatzikyriakidis and Luo 2015))
- Introduction of intenTional contexts: Contexts including the intentions (rather than the beliefs) of an agent. We can use this idea for adverbs like intentionally:
  - $[\![Intentionally]\!] = \lambda x : [\![human]\!] .\lambda P : [\![human]\!] \rightarrow Prop.\ I_x(P(x)) \wedge \Gamma(P(x))$
- Domain adverbs, e.g. *botanically, mathematically*
  - $[\![botanically]\!] = \lambda P : Prop.\Gamma_B P$
- Intensionality without possible worlds

MTT semantics in Coq

- Encoding MTT semantics based on theoretical work using Type Theory with Coercive Subtyping in Coq
  - Coq is a natural toolkit to perform such a task
    - The type theory implemented in Coq is quite close to Type Theory with Coercive Subtyping
    - Thus, the TT does not need to be implemented!
    - What we need, is a way to encode the various assumptions as regards linguistic semantics and then reason about them

Reasoning with NL

- As soon as NL categories are defined, Coq can be used to reason about them
  - In effect, we can view a valid NLI as a theorem
    - ★ Thus, we formulate NLIs as theorems
    - ★ The antecedent and consequent must be of type *Prop* in order to be used in proof mode
    - ★ Thus, the first can be formulated as a theorem, but not the second:

```
Theorem EX:(walk) John-> some Man (walk).
Theorem WA:walk -> drive.
```

Reasoning with NL

- The same tactics that can be used in proving mathematical theorems are used for NL reasoning
  - The aim is to predict correct NLIs while avoiding unwanted inferences
    - For example, given the semantics for quantifier *some*, one can formulate the following theorem and further try to prove it

```
Theorem EX:  (walk) John-> some Man (walk).
```

An NLI example

- Basically, from a sentence like *John walks*, we should infer that *a man walks*
- We formulate the theorem

```
Theorem EX:  (walk) John-> some Man (walk).
```

- We unfold the definition for *some* and use *intro*

```
EX < intro.
1 subgoal
H : walk John
============================
exists x : Man, walk x
```

- We use the exists tactic to substitute *x* for *John*. Using *assumption* the theorem is proven

An NLI example

- To the contrary, we should not be able to prove the opposite

```
Theorem EX: some Man (walk) -> (walk) John.
```

- Indeed, no proof can be found in this case.
  - We unfold *some* and use *intro*

```
EX < intro.
1 subgoal
H : exists x : Man, walk x
============================
walk John
```

  - From this point on, we can use any of the *elim*, *induction*, *case* tactics but at the end we reach a dead end

```
EX < intro.
1 subgoal
H : exists x : Man, walk x
x : Man
H0 : walk x
============================
walk John
```

The FraCas test suite

- As already said, the examples involve a number of premises, followed by a question (h).
  - We reformulate the examples as involving declarative forms in Coq (this is a usual approach, at least with deep approaches)
    - In cases of *yes* in the FraCas test suite, we formulate a declarative hypothesis as following from the premise
    - In cases of *no*, we formulate the negation of a declarative hypothesis as following from the premise
    - In cases of *UNK*, for both the positive and the negated h, no proof should be found. If it is, we overgenerate inferences we do no want

The FraCas test suite

- Quantifier monotonicity

  (9) Some Irish delegates finished the survey on time
  Did any delegates finish the survey on time? [YES]

  - Standard semantics for indefinites *some* and *any* (no presuppositions encoded)

```
Definition some:= fun A:CN, fun P:A->Prop=> exists x:A,
P(x).
```

- Σ types as dependent record types

```
Record Irishdelegate:CN:=mkIrishdelegate{c:> Man;_:
Irish c}.
```

- These assumptions suffice, the subtyping relation via π_1 does the trick here

```
Theorem IRISH: (some Irishdelegate)(On_time(finish(the
survey)))->(some delegate)(On_time (finish(the survey))).
```

Quantifier monotonicity

- Monotonicity on the second argument

  (10) Some delegates finished the survey on time
  Did some delegates finish the survey? [YES, FraCas 3.71]

- We define the auxiliary object and then *on_time*

```
Parameter ADV: forall (A : CN) (v : A -> Prop),sigT
(fun p : A -> Prop => forall x : A, p x -> v x).
Definition on_time:= fun A:CN, fun v:A->Prop=> projT1
(ADV(v)).
```

- These assumptions suffice for these cases

```
IRISH2 < Theorem IRISH2: (some delegate)(on_time
(finish(the survey)))->(some delegate)((finish
(the survey))).
```

Adjectives

- Affirmative adjectives (these are intersective adjectives)

  (11) John has a genuine diamond
  Does John have a diamond? [Yes, FraCas 3.197]

  - The Σ type approach will work here
- Opposites
  - Here we need to get:

  (12) $\mathsf{Small}(N) \Rightarrow \neg\, \mathsf{Large}(N).$
  $\mathsf{Large}(N) \Rightarrow \neg\, \mathsf{Small}(N)$
  $\neg\, \mathsf{Small}(N) \not\Rightarrow \mathsf{Large}(N).$
  $\neg\, \mathsf{Large}(N) \not\Rightarrow \mathsf{Small}(N)$

- The problem is that there are other sizes than a binary opposition *small-large*, e.g. normalsized items
- Use this intuition:

```
Definition small:= fun A:CN, fun a:A=> not (large (a)
/\ not (normalsized (a)).
```

Adjectives

- Some examples that are correctly predicted with the definition given

(13) Mickey is a small animal
Is Mickey a large animal? [No, FraCas 3.204)

(14) Fido is not a large animal
Is Fido a small animal? [UNK, FraCas 3.207)

(15) All mice are small animals
Mickey is a large mouse
Is Mickey a large animal? [No, FraCas 3.210)

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